NAYPYITAW STATE POLYTECHNIC UNIVERSITY(NSPU)

Department of Computational Mathematics

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Engineering Mathematics IV
(Fourier Analysis and Complex Analysis)
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Objectives,

- ➢**To understand Exponential Function**
- ➢**To understand Trigonometric and Hyperbolic Functions**
- ➢To apply problems by Theory
- ➢To solve problems by Theory

Exponential Function

One of the most important analytic functions, the complex exponential function is written e^z , also written exp z, **Properties**:

(1) $e^z = e^{x+iy} = e^x . e^{iy} = e^x(cosy + isiny)$ $e^z = e^x$for real $z = x$ because $cosy = 1$ and $siny = 0$ when $y = 0$, e^z is analytic for all z.

(2) $(e^z)' = ez$, a function *f(z)* that is analytic for all z is called an entire function.

(3) $e^{z_1+z_2} = e^{z_1}e^{z_2}$.

(4)$e^z = e^x . e^{iy}$,Furthermore, for $z = iy$ we have from(1) the so-called Euler formula,

(5) $e^{iy} = \cos y + i\ siny\ and\ e^{-iy} = \cos y - i\ siny$ is

Polar form of complex number : $z = r\ (\cos\theta + i\sin\theta)$,

(6)$z = r\ e^{i\theta}$ (7)$e^{2\pi i} = 1$

(8) $e^{2\pi i} = cos2\pi + i\ sin2\pi = 1 + 0 = 1, e^{\pi i} = cos\pi + i\ sin\pi = -1 + 0 = -1$,

$e^{-\frac{\pi}{2}i} = cos\frac{\pi}{2} - i\ sin\frac{\pi}{2} = 0 - i = i, \quad e^{-\pi i} = cos\pi - i\ sin\pi = -1 + 0 = -1$

$(9)\left|e^{iy}\right| = |\cos y \ + \ i\ siny| = \sqrt{cos^2y + sin^2y} = 1$

$(10)\ |e^z| = \left|e^{x+iy}\right| = \left|e^x . e^{iy}\right| = |e^x|\left|e^{iy}\right| = \ e^x, \arg e^z \ = \ y \ \pm \ 2n\pi \ \ (n \ = \ 0,1,2, \dots\ )$.

$(11)e^z \neq \ 0\ for\ all\ z, (12)e^{z+2\pi i} = \ \ e^z\ \ for\ all\ z, . (13)\ -\pi < y \leq$

This infinite strip is called a fundamental region of e^z .

Example 1. Function Values. Solution of Functions

Compute $e^{1.4-0.6i}$, $\left|e^{1.4-0.6i}\right|$ $and\ Arg\ e^{1.4-0.6i}$.

Solution:

$e^{1.4-0.6i} = e^{1.4}.e^{-0.6i}$

$= e^{1.4}\,(\cos 0.6 \ - \ isin 0.6)\,, = \ e^{1.4}\cos 0.6 \ - \ ie^{1.4}\sin 0.6\,, = \ 3.35 - \ i2.29$

$\left|e^{1.4-0.6i}\right| = |e^{1.4}.e^{-0.6}| = |e^{1.4}| = e^{1.4} = 4.055\,, Arge^{1.4-0.6i} = \ -0.6$

Compute e^{2+i}, e^{4-i}

Solution: $e^{2+i} = e^2\,(\cos 1 \ + \ isin 1), e^{4-i} = e^4\,(\cos 1 - i \sin\ 1), = e^2(\cos 1 + isin 1)$

$e^4\,(\cos 1 - i \sin\ 1)\ ,= e^2\ e^4\,(\cos^2 1 + \sin^2\ 1) = e^6$

Compute $e^z = 3 + 4i$,

Solution: $|e^z| = e^x = \sqrt{3^2 + 4^2} = 5, e^x = 5$

$x = \ln(5) = 1.609, e^z = e^x(cosy + i\ siny), e^x \cos y = 3, \quad e^x \sin y = 4$

$\cos y = 0.6, \quad \sin y = 0.8, \quad y = 0.927$

$\therefore z = x + iy = 1.609 + 0.927i \pm 2n\pi i \quad (n = 0,1,2, \dots)$

Trigonometric and Hyperbolic Functions

Euler Formulas ;$e^{ix} = \cos x + i\sin x, \quad e^{-ix} = \cos x - i\sin x$

By addition and subtraction , $e^{ix} + e^{-ix} = (cosx + isinx) + (cosx - isinx) = 2cosx,$

$\therefore cosx = \frac{1}{2}(e^{ix} + e^{-ix}), \therefore cosz = \frac{1}{2}(e^{iz} + e^{-iz}),$

$e^{ix} - e^{-ix} = (cosx + isinx) - (cosx - isinx) = 2isinx$

$\therefore sinx = \frac{1}{2i}(e^{ix} - e^{-ix}), \therefore sinz = \frac{1}{2i}(e^{iz} + e^{-iz}),$

General Formulas

$$\cos(z_1 \pm z_2) = cosz_1cosz_2 \mp sinz_1sinz_2, \sin(z_1 \pm z_2) = sinz_1cosz_2 \pm cosz_1sinz_2$$

$cos^2 z + sin^2 z = 1,\ cos^2 z - sin^2 z = \cos 2z$

Hyperbolic Functions

The complex hyperbolic cosine and sine, $\cosh z = \frac{1}{2}(e^z + e^{-z}), \sinh z = \frac{1}{2}(e^z - e^{-z})$

$$\frac{d}{dz}\cosh z = \sinh z, \qquad \frac{d}{dz}\sinh z = \cosh z$$

Relation between Complex Trigonometric and Hyperbolic functions

$\cosh iz = \cos z, \quad sinh\ iz = i\sin z, \cos iz = \cosh z, \sin iz = i\sinh z$

Example 1. Real and Imaginary Parts. Absolute Value. Periodicity

Show that ;(a) $\cos z = \cos x \cosh y - i \sin x \sinh y$ (b) $\sin z = \sin x \cosh y + i \cos x \sinh y$

Proof (a) $\cos z = \frac{1}{2}\left(e^{iz} + e^{-iz}\right) = \frac{1}{2}\left(e^{i(x+iy)} + e^{-i(x+iy)}\right) = \frac{1}{2}\left(e^{ix-y} + e^{-ix+y}\right)$

$= \frac{1}{2}e^{-y}(\cos x + i\sin x) + \frac{1}{2}e^{y}(\cos x - i\sin x) = \frac{1}{2}(e^y + e^{-y})\cos x + \frac{1}{2}\, i\,(-e^y + e^{-y})\sin x = \cos x \cosh y - \frac{1}{2}(e^y - e^{-y}) i \sin x = \cos x\ coshy - isinx\ sinhy$

Proof (b): $\sin z = \frac{1}{2i}\left(e^{iz} - e^{-iz}\right) = \frac{1}{2i}\left(e^{i(x+iy)} - e^{-i(x+iy)}\right) = \frac{1}{2i}\left(e^{ix-y} - e^{-ix+y}\right)$

$$= \frac{1}{2i}\left(e^{ix}.e^{-y} - e^{-ix}.e^{y}\right) = \frac{1}{2i}\left(e^{-y}(\cos x + i\sin x) - e^{y}(\cos x - i\sin x)\right)$$

$$= \frac{1}{2i}\left(e^{-y} - e^{y}\right)\cos x + \frac{1}{2}\left(e^{y} + e^{-y}\right)\sin x$$

$$= \frac{1}{2i}(-1)\left(e^{y} - e^{-y}\right)\cos x + \sin x\cosh y$$

$$= \frac{1}{2i}(i^2)\left(e^{y} - e^{-y}\right)\cos x + \sin x\cosh y = i\cos x\sinh y + \sin x\cosh y$$

$$= \sin x\cosh y + i\cos x\sinh y$$

Example 2. Solution of Equations. Zeros of *cosz* and *sinz*

Solve (a) $\cos z = 5$ (which has no real solution!) ,(b) $\cos z = 0$ (c)$\sin z = 0$.

Solution : (a) $e^{2iz} - 10e^{iz} + 1 = 0$ from (1) by multiplication by e^{iz} .

This is a quadratic equation in e^{iz}, with solutions (rounded off to 3 decimals)

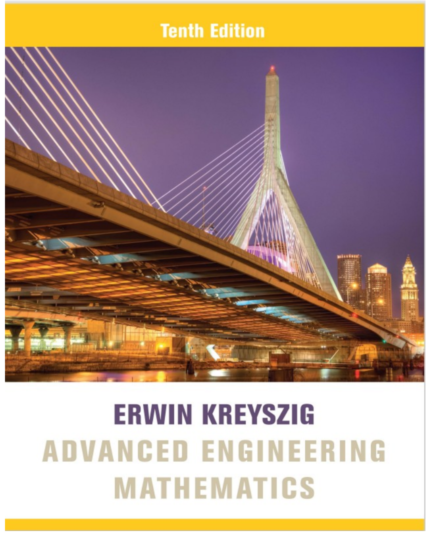
$e^{iz} = e^{-y+ix} = 5 \pm \sqrt{25-1} = 9.899$ and 0.101.

$e^{iy} = 9.899\ or\ 0.101, e^{ix} = 1, y = \pm 2.202, x = 2n\pi$

$Ans: z = \pm 2n\pi \pm 2.292i\ (n = 0,1,2 \dots)$

(b)$\cos x = 0, \sinh y = 0\ by\ (7\ a), y = 0.$ $Ans: z = \pm \frac{1}{2}(2n + i)\pi\ (n = 0,1,2 \dots)$

(c)$\sin x = 0, \sin h\ y = 0\ by\ (7\ b), y = 0.$ $Ans: z = \pm n\pi\ (n = 0,1,2 \dots)$



Reference Book

- Advanced Engineering Mathematics, Erwin Kreyszig, 10^{th} Edition

Thank you